

Bulawayo Waste Hub: User Manual

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About this demo

Bulawayo Waste Hub is a clickable demonstration of how residents, the City Council, EMA, churches, residents' associations, business and waste pickers could share one system to report, fix and track dumped waste, ward by ward. It supports Karl Malotane's PhD study on a multi-stakeholder framework for waste management in Bulawayo.

Everything in it is invented sample data. It is not connected to the City Council, EMA, WhatsApp or any phone network, and nothing you do in it reaches anyone else.

Where to open it: bulawayo-waste-hub.vercel.app on any phone, tablet or computer, with taffn.github.io/bulawayo-waste-hub as a backup link. Both show the same demo. No account or installation is needed.

This manual covers all six stages of the finished demo. Each stage section explains the screen and ends with a short checklist you can follow to test it.

Introduction: how the app serves the study

The app is the study's draft framework made visible: each screen puts one part of the proposed multi-stakeholder framework into daily practice, so stakeholders can see how it would work before the survey tests it.

The proposal argues that Bulawayo's waste problem is mainly a coordination problem, and that a lasting waste culture needs reliable service, legitimate institutions, financial trust and tools that include informal actors. The app answers that diagnosis directly:

Study element	How the app expresses it
RQ1: collaborative leadership and behaviour change	Partner queues, routing rules and the ward review give every partner a defined role, a deadline and a shared record of what was agreed and done
RQ2: why clean-up campaigns have not lasted	Residents can report on any day, not only clean-up Friday, and each report is followed to a fix and a site check, closing the gap between campaign days

Study element	How the app expresses it
RQ3: a practical, acceptable framework	The Charter (who handles what, and how fast) is a setting the partners agree and sign, not code, so it can be negotiated and changed
H3: including waste pickers and private firms	Pickers register by basic phone, operators receive cases, and buyers confirm recovered material
Questionnaire Sections B-E	The indicators in Settings mirror the questionnaire, with empty baseline slots for the survey results
Inclusive tools (proposal section 2.5.4)	WhatsApp and a no-data USSD menu, in English, isiNdebele and chiShona

Timing. The demo is shown now, before the survey, as a picture of how the framework could run. When the results are in, they are loaded into the app as baselines, roles and routing rules, and features the findings do not support are removed. Real use by residents waits until after data collection, so it does not influence what the survey measures.

Getting started

Two controls at the top of every screen decide what you see: **Viewing as** (your role) and **Ward** (Cowdray Park, or the placeholder Ward B). The menu on the left, or across the top on a phone, opens each screen; the demo opens on Start here.

Viewing as	What changes
Ward committee	Sees every report, owner and due time; sees every partner's queue
Resident	Sees reports and their status, but not who is assigned; has no queue
City Council (BCC), EMA, Private operator, Ward volunteer team, Waste picker	Their own cases are highlighted on the map; Partner queues shows only their own list
Residents' association	Partner queues shows fixed cases waiting to be checked on site
Church, Business buyer	No cases to fix; their tasks arrive in later stages

Viewing as	What changes
Administrator	Also opens Data tables and Settings

Starting over: choose Administrator, open Settings, tap **Reset demo data**, then **Reset**. This restores the original sample reports and clears both demo phones. Your changes are only ever saved in your own browser, so each person who opens the link starts with a clean demo.

Stage 1: Ward map and reports

The **Ward map** shows every report in the chosen ward as a pin on a schematic map (not to scale).

- **Pin colour** is the status: blue new, amber assigned, purple in progress, green fixed.
- **Pin size** is the size of the dump; a **red ring** means overdue; a **small number** means several reports of the same site were merged.
- **Status chips** above the map filter the pins; tap "All open and fixed" to clear the filter.
- **Tap a pin** to open the report: where, size, owner, due time, photo, AI tags and a full history of who did what.

Reports lists the same reports as a table, newest first; tap a row to jump to it on the map. **Data tables** (Administrator) shows the underlying database table by table, and **Settings** shows the routing rules, indicators awaiting survey results, and decisions still pending.

Test it

- ☐ The map loads with pins and the chips show counts
- ☐ Tapping the large pin by the market shows a report with merged duplicates
- ☐ The Overdue chip leaves only red-ringed pins
- ☐ Viewing as City Council fades other partners' pins; Resident hides owner and due time
- ☐ Ward B shows the dashed placeholder area
- ☐ Administrator can open Data tables and Settings

Stage 2: Reporting by WhatsApp and USSD

The **WhatsApp and USSD** screen holds two simulated phones and a log called "What the system did". A report sent from either phone is tagged by AI, checked against open reports at the same spot, and then either merged with that case or opened as a new case and routed to a partner.

WhatsApp (smartphone)

1. Choose a language: English, isiNdebele or chiShona.
2. Tap "Report dumped waste" and pick a sample photo, or attach your own with **Photo**.
3. Confirm or change the waste type the AI suggests, then choose the size, the place and any hazard.
4. Tap **Send report**. The bot replies with a reference number and says who will deal with it.
5. "Check a report" takes a reference such as 1041; "Collection days" lists the ward's schedule.

USSD (basic phone, no data needed)

1. Tap **Type *123*77#** (a fictional code), then **Send**.
2. Choose a language, then answer each numbered menu with the keypad and **Send**.
3. Option 3 registers a waste picker, who appears in Data tables under pickers.

The isiNdebele and chiShona wording is a draft awaiting native-speaker review. The AI's suggested waste type comes from the sample photo; it cannot read your own photo yet.

Test it

- ☐ A WhatsApp report at the market stalls is merged into the existing case
- ☐ A report somewhere new opens a case routed to a partner, with a dotted pin on the map
- ☐ A USSD report works in all three languages
- ☐ "Check a report" returns the status of 1041

Stage 3: Partner queues

Partner queues is where partners work their cases, overdue first. What you see depends on Viewing as.

- **Ward committee or Administrator:** a tab for each partner, plus "Not yet routed" and "Awaiting verification". **Route by the Charter** sends a new report to the right partner; **Route all** does every new report at once.
- **A partner (BCC, EMA, operator, pickers, ward team):** only that partner's open cases, with recently fixed ones below.
- **Residents' association:** fixed cases waiting for a site check.

Working a case

1. **Start work** moves the case to In progress.

2. **Mark fixed**, choose an after photo (the sample "Site cleared" or your own), then **Confirm fixed**. Everyone who reported the site is told: a WhatsApp message on the demo smartphone, an SMS on the demo basic phone.
3. **Reassign** moves the case to another partner, for example from BCC to the private operator when a truck is down.
4. The residents' association then taps **Confirm cleared**, or **Still there**, which reopens the case and gives the owner 24 hours.

Every step is written into the case history on the map, with who did it and when. Fixed cases show before and after photos side by side.

Test it

- ☐ As BCC, a new WhatsApp report appears in the queue and can be started and fixed
- ☐ The "fixed" message appears in the WhatsApp chat and as an SMS on the basic phone
- ☐ As Residents' association, "Still there" sends the case back to BCC
- ☐ Reassign moves a case to another partner's tab

Stage 4: Pickers and recyclables

Pickers and recyclables recognises informal waste pickers as partners: they are registered, log the material they recover, and are paid by a registered buyer once the weight is confirmed. The top row shows registered pickers, kilograms confirmed in the last 30 days, money paid, and purchases waiting.

1. **Register a picker** (Ward committee, Residents' association, Waste picker or Administrator): choose the main material and phone type, then **Register picker**. Pickers can also register themselves on the basic phone, option 3.
2. **Log recovered material**: choose the picker, material and weight, and optionally the reported site it came from, then **Log for buyer to confirm**. A linked site gets a "Recovered" entry in its history.
3. **Confirm the purchase** (Business buyer): tap **Confirm and pay**, or **Different weight** to record the buyer's scale reading instead. Payment is simulated mobile money; a picker registered on the demo basic phone receives an SMS.

Prices per kilogram are invented examples for the demo. The totals feed indicator C2 and hypothesis H3 on including waste pickers.

Test it

- ☐ Register a picker on the basic phone and see them in the register
- ☐ As Waste picker, log 30 kg of PET plastic for that picker
- ☐ As Business buyer, confirm it at a different weight and see the payment

- ☐ The basic phone shows the payment SMS

Stage 5: Review and forum

Review and forum turns the month's activity into the evidence partners meet around. It has two tabs.

Ward review pack is prepared for the ward committee after each clean-up day and covers the last 30 days:

- **Headline figures:** reports received, the share made between clean-up days, median time to fix, cases overdue now, and kilograms recovered by pickers.
- **AI summary:** a plain-language paragraph drafted from the ward's data, which the committee edits before sharing.
- **How each partner did:** new cases, fixed, median fix time, share fixed on time, open and overdue, for every partner.
- **Repeat sites:** places reported again and again, which need a standing fix rather than another clear-up.
- **Last review and Action tracker:** who attended, the actions agreed with owner and due date, and their status. Tap **Record today's review** to start a new review, add actions with **Add action**, and change a status from its dropdown. Late actions show as Overdue.

Forum overview is for the City Waste Partnership Forum: every ward side by side, and the framework's indicators. Questionnaire indicators (Sections B-E) show "Awaiting survey" until the baseline is loaded; app indicators such as median fix time update daily; targets are left for the forum to set.

Test it

- ☐ The review pack shows figures, an AI summary and a partner table for Cowdray Park
- ☐ Adding an action puts it in the tracker with the chosen owner and due date
- ☐ Marking the overdue patrol action Done updates the summary
- ☐ Forum overview lists both wards and the indicators with survey slots

Stage 6: Demo guide and feedback

Start here is the first screen. It explains the system in one paragraph and gives a nine-step, ten-minute demo script for stakeholder meetings. Each step says what to show, why it matters for the framework, and which role to use; its **Go** button opens the right screen with the right role.

Feedback collects stakeholder views in two minutes: who they are, four 1–5 scores (easy to understand, helps coordination, fair to pickers and residents, my organisation would use it), and what is missing or worrying.

- Pass one device round a meeting: each person taps **Save my feedback** and hands it on.
- Answers stay on that device, separate from the demo data, so **Reset demo data** does not delete them.
- Karl taps **Copy all as spreadsheet** and pastes the answers into Excel or Google Sheets. Each person can also copy their own answers or share them on WhatsApp.

Test it

- ☐ The demo opens on Start here, and each Go button opens the right screen and role
- ☐ Saving feedback without a score asks for one; with scores it is saved
- ☐ Copy all as spreadsheet gives one row per answer

Running a stakeholder demo

1. Before the meeting: open the link, choose Administrator → Settings → **Reset demo data**.
2. Open **Start here** and follow the nine steps, using each **Go** button.
3. End on **Feedback** and pass the device round.
4. Afterwards: **Copy all as spreadsheet** and save the answers with the meeting notes.

Troubleshooting and common questions

- **Someone else can't see my test reports.** Expected: each browser keeps its own copy of the demo, and nothing is shared between viewers.
- **The demo looks cluttered after a lot of testing.** Administrator → Settings → Reset demo data → Reset.
- **My own photo disappeared.** Uploaded photos are kept only until the page reloads; sample photos always stay.
- **Feedback answers are missing.** They are kept on the device where they were entered; open the demo on that same device and browser to copy them out.
- **Basic-phone place names end in "...".** Long names are shortened to fit the small screen; shorter local-language wording will fix this.
- **The link shows an old version.** Refresh the page; GitHub can take a few minutes to publish an update.

Conclusion

The finished demo shows, end to end, how the study's multi-stakeholder framework could run in a Bulawayo ward. A resident reports on any day, in their own language, from any phone; the Charter sends the case to the partner responsible; the partner fixes it against a deadline; residents check the site; waste pickers are recognised and paid; and every month the ward committee and the city forum review the same evidence and agree what to do next.

Each of those steps puts one part of the draft framework into practice: shared leadership through clear roles and joint review, collaboration mechanisms that include pickers, women and residents, and a waste culture measured by what happens between clean-up days rather than on them. The indicators already mirror the questionnaire, so the survey results can be loaded in as baselines when Karl's data collection is complete.

What comes next

1. Show the demo to stakeholders with the demo guide and gather their feedback.
2. Have native speakers check the isiNdebele and chiShona wording, and choose the second pilot ward.
3. After the survey, load the baselines and adjust roles, routing rules and indicators to the Chapter Five framework.
4. With stakeholder agreement, build the live version: real WhatsApp and USSD channels, a shared database, a live AI model and a POTRAZ data-controller licence, then pilot it in two wards as the living model.

Everything in this demo is sample data. The live version must be agreed with the partners and licensed before any resident uses it.